

Sapphire Coding and Usage Guide

Language syntax, standard-library namespaces, CLI usage, and verified boundaries.

This manual describes the current local implementation. Feature names are not evidence of production scale, hardware availability, or successful external actions.

Language

The tree-walk interpreter supports bindings, functions, control flow, arrays, maps, structs, lambdas, exceptions, pipes, process literals, and parallel blocks.

Namespaces

The runtime exposes `os`, `fs`, `http`, `ai`, `data`, `scheduler`, `ml`, `gui`, and `agent` namespaces. Optional packages and services are required for some operations.

Execution

Run `.sp` files with `sapphire run`, use the REPL or `eval` command, and inspect source through the compiler tools. The compiler GUI uses the interpreter and textual IR; it is not a native compiler.

Example

```
let info = os.system_info(); print("CPU: {info.cpu_usage_percent}%");
```

Version 1.0.5. Reproduce local measurements with `benchmarks/benchmark_runtime.py`. Values depend on the machine and environment.